

PCD ASSIGNMENT 01

Down Sampling and Up Sampling

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1. Introduction

Digital images consist of pixels that represent the visual information of an image. Changing the number of pixels can affect the resolution and the amount of detail that can be represented. In this experiment, downsampling and upsampling are performed to observe these changes and compare several methods.

Four images with different characteristics are used as input: cat.jpg, owl.jpg, gelex.jpg, and photostrip.jpg. The images contain different types of details, textures, colors, and edges, which makes them suitable for comparing the effects of the sampling methods.

The experiment consists of two main parts. The first part is downsampling using Max, Average, and Median methods. The second part is upsampling using Nearest Neighbor, Bilinear, and Bicubic interpolation. An additional reconstruction experiment is also performed by downsampling the images using Average and then returning them to their original resolution using the three interpolation methods.

2. Downsampling

Downsampling is performed to reduce the spatial resolution of an image. In this experiment, a factor of 2 is used. Each 2×2 block of pixels from the original image is represented by one pixel in the downsampled image.

Max

Max downsampling selects the highest intensity value from each 2×2 block. Because the highest value is selected, brighter pixels have a stronger influence on the output.

In the cat image, this can be seen in areas such as the white paws and white parts of the face. Bright areas tend to remain prominent after downsampling. This method can preserve bright details, but it can also make some bright regions appear more dominant than they are in the original image.

Average

Average downsampling calculates the mean value of the pixels within each 2×2 block. This produces a smoother representation because the intensity values are combined instead of selecting only the brightest or darkest pixel.

For the cat and owl images, the texture of the fur and feathers becomes smoother, but some of the smaller details are reduced. In the photostrip image, small details and text are also affected because several pixels are combined into a single pixel.

Median

Median downsampling selects the median intensity value from each 2×2 block. This gives a representative value for the local area without giving as much importance to an unusually high or low pixel.

The Median result is relatively balanced compared with Max and Average. The difference is more noticeable in areas where the image contains strong intensity changes or detailed textures.

Overall, the downsampling results show that reducing the resolution also reduces some fine details. The effect is more noticeable in images containing textures, sharp edges, and small objects.

3. Upsampling

Upsampling is used to increase the spatial resolution of an image. In this experiment, the images are enlarged by a factor of 2 using Nearest Neighbor, Bilinear, and Bicubic interpolation.

Nearest Neighbor

Nearest Neighbor uses the value of the closest existing pixel to determine the value of a new pixel. This makes it the simplest of the three interpolation methods.

The main effect can be seen as a blocky or pixelated appearance. This is especially noticeable around detailed areas such as the cat's fur, the owl's feathers, and the small details in the photostrip.

Bilinear

Bilinear interpolation calculates new pixel values using surrounding pixels. This produces smoother transitions compared with Nearest Neighbor.

The enlarged images therefore look less pixelated. However, the smoothing effect can reduce the sharpness of some details. For example, the texture of the cat's fur and the owl's feathers appears softer than in the original image.

Bicubic

Bicubic interpolation uses a larger neighborhood of pixels to estimate the values of new pixels. The result generally appears smoother and more natural than Nearest Neighbor.

In the experiment, Bicubic provides a smoother appearance around edges and detailed regions. It maintains the overall appearance of the images while avoiding the strong blocky effect produced by Nearest Neighbor.

4. Downsampling and Reconstruction

To further examine the effect of downsampling, the images that were reduced using the Average method are enlarged back to their original resolution using Nearest Neighbor, Bilinear, and Bicubic interpolation.

The results show that some information is lost when the image is downsampled. When it is enlarged again, the interpolation methods can estimate new pixel values, but they cannot completely recover the original details.

The Average + Nearest Neighbor result shows the most visible pixelation. Average + Bilinear produces a smoother result, although some details appear blurred. Average + Bicubic produces a smoother reconstruction and generally gives the most natural-looking result among the three.

This effect is particularly visible in areas with fine details. The fur in the cat image, feathers in the owl image, detailed areas in the Gelex image, and small text and photographs in the photostrip all show some loss of detail after the downsampling and reconstruction process.

5. Comparison Between the Images

The four images produce different results because their visual characteristics are different.

The **cat image** is a grayscale image with detailed fur and clear edges. The changes caused by downsampling and upsampling are therefore easy to observe in the fur, face, and paws.

The **owl image** contains detailed feathers as well as a complex background. Downsampling reduces some of the smaller feather details, while the different interpolation methods produce different levels of smoothness when the image is enlarged.

The **Gelex image** contains many colors, bright areas, and a detailed background. Max downsampling makes bright areas more dominant, while Average produces smoother transitions between different regions.

The **photostrip image** contains photographs, faces, text, and other small details. The reduction in resolution makes small text and fine details less distinct, making this image useful for observing information loss.

Using these four images makes it possible to see that the effects of sampling methods are not exactly the same for every type of image.

6. Overall Analysis

Based on the experiment, Max, Average, and Median produce different results when reducing the image resolution. Max emphasizes the highest intensity values, Average produces smoother intensity transitions, and Median provides a representative value for each local region.

For upsampling, Nearest Neighbor produces the most noticeable pixelation because it directly uses the closest pixel value. Bilinear produces smoother transitions but can make details slightly softer. Bicubic produces a smoother reconstruction and generally gives a more natural appearance.

The comparison also shows that image characteristics affect how noticeable these changes are. Images containing fine textures, sharp edges, small text, and strong intensity differences show more visible changes after their resolution is reduced.

The reconstruction experiment further demonstrates that increasing the resolution does not mean that lost information is recovered. Interpolation creates estimated pixel values based on the available information, so details removed during downsampling cannot be completely restored.

7. Conclusion

The experiment demonstrates the effects of different downsampling and upsampling methods on four images with different characteristics.

For downsampling, Max emphasizes bright pixels, Average creates smoother intensity values, and Median provides a representative local value. For upsampling, Nearest Neighbor produces a more pixelated result, while Bilinear and Bicubic produce progressively smoother results.

The results also demonstrate the relationship between spatial resolution and image detail. Reducing the number of pixels causes information to be lost, particularly in areas containing fine textures and small details. When the image is enlarged again, interpolation can estimate new pixels, but it cannot fully recover information that was already lost.

Overall, the experiment shows that the choice of sampling method affects the appearance and quality of the resulting image, and the most suitable method depends on the characteristics of the image and the desired result.